

Module 13911 - Algebra: Structures and Algorithms - Lecture

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0 The Very Basics

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The key structures in algebra: groups, rings and fields.

Definition (Abelian) Group

An *abelian group* is a set G with an operation $*$: $G \times G \rightarrow G$ which satisfies the following $\forall a, b, c \in G$:

$$* : \mathbb{R} \setminus \{0\} \times \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R} \setminus \{0\}$$
$$a * b = b * a \quad (\text{C})$$

$$a * (b * c) = (a * b) * c \quad (\text{A})$$

$$a * e = a \quad (\text{N})$$

$$a * (a^{-1}) = e \quad (\text{I})$$

$\Rightarrow$ without the commutativity G would just be called a *group*.

$\rightarrow$ Group Theory (Abelian Groups are much easier than general ones)

Definition Ring

A *ring* is a set R with the operations $+$: $R \times R \rightarrow R$ and $\cdot$: $R \times R \rightarrow R$ which satisfies the following $\forall a, b, c \in R$:

CA : $a + b = b + a, \forall a, b \in R$

AA : $a + (b + c) = (a + b) + c, \forall a, b, c \in R$

NA : There exists a unique element in R , denoted by 0 or 0_R , such that $a + 0 = a, \forall a \in R$

IA : For every $a \in R$ there exists a unique element $-a \in R$ such that $a + (-a) = 0$

AM : $a \cdot (b \cdot c) = (a \cdot b) \cdot c, \forall a, b, c \in R$

D : $a \cdot (b + c) = a \cdot b + a \cdot c, \forall a, b, c \in R$ and $(a + b) \cdot c = a \cdot c + b \cdot c, \forall a, b, c \in R$

$\rightarrow$ A ring $(R, +, \cdot)$ is called *commutative* when:

CM : $a \cdot b = b \cdot a, \forall a, b \in R$

$\rightarrow$ A ring $(R, +, \cdot)$ is called a *unitary* when:

NM : There exists a unique element in R , denoted by 1 or 1_R , such that $a \cdot 1 = a, \forall a \in R$

$\rightarrow$ A ring $(R, +, \cdot)$ is called *proper* if in that ring one has $0_R \neq 1_R$.

$\rightarrow$ Theory of commutative rings is called commutative algebra.

Definition Field

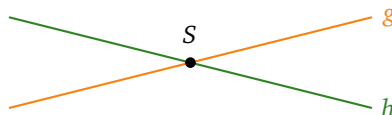
A proper, unitary, commutative ring $(R, +, \cdot)$ is called a *field* if all non-zero elements of R have a multiplicative inverse.

$\rightarrow$ Galois Theory, Theory of finite fields, algebraic closure, etc.

1 Algebra, Geometry and Algorithms

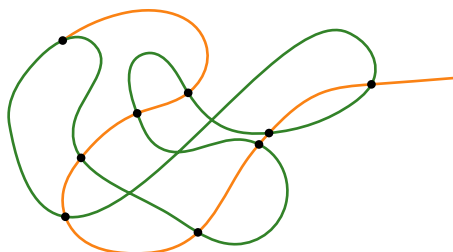
Linear Algebra

- **Algebra:** linear equations
- **Geometry:** sets defined by linear subsystems of equations, affine subspaces, vectors, vector subspaces, ...
- **Algorithms:** Gaussian elimination



Non-Linear Algebra

- **Algebra:** polynomial equations
- **Geometry:** Varieties (sets defined by systems of polynomial equations)
- **Algorithms:** Buchberger's algorithm (a far-reaching generalization of Gaussian elimination), Gröbner bases



The more Technical Layer

Notation of ideal in a ring

→ this notation is relevant for the above topics but also for construction of rings and fields
(the latter is used in coding theory and cryptography)

1.1 Polynomials and the Affine Space

Variables considered per default in algebra are *formal variables (symbols)*. Formal means roughly placeholder for anything. There is no domain attached to this.

Formal variables are used in *formal expressions (symbolic expressions)*.

Note the difference to functions: Symbolic expressions are just expressions, not functions. In particular, they don't have any domain or co-domain.

In what follows, we consider n formal variables ($n \in \mathbb{Z}_{>0}$) whose default names will be $x_1, \dots, x_n$. If $n = 1$ it's customary to use x and y . For $n = 3$ it's customary to use x, y, z .

Definition 1.1.1

Let $x = (x_1, \dots, x_n)$ be the vector of n formal variables and $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{Z}_{\geq 0}^n$ a vector of n non-negative integers. Then the expression

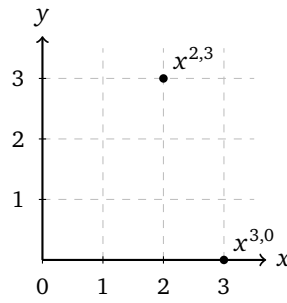
$$x^\alpha := x_1^{\alpha_1} \cdot \dots \cdot x_n^{\alpha_n}$$

is called the *monomial for the exponent vector α* . $\alpha_1 + \dots + \alpha_n$ is called the *(total) degree of the monomial x^α* , denoted as $|\alpha|$.

Our *coefficient domain* will be an arbitrary field which we denote by k throughout.

Example

Given: $x^{2,3} = x_1^2 \cdot x_2^3$ and $x^{3,0} = x_1^3$



You already know some fields: $\mathbb{Q}, \mathbb{R}, \mathbb{C}\mathbb{F}_p$ and $\mathbb{F}_{p^N}$ where p is prime, $\mathbb{Q}[\sqrt{2}]$

→ It's important to know some examples of them.

Definition 1.1.2

A polynomial in variables $x_1, \dots, x_n$ with coefficients in k is a formal expression of the form

$$f = \sum_{\alpha} c_{\alpha} x^{\alpha}, \quad (1)$$

where α runs over $\mathbb{Z}_{\geq 0}^n$ and $c_{\alpha} \neq 0$ for any finitely many choices of k .

(You must have seen polynomials in one variable before... surprise, surprise - life is not one-dimensional!)

$c_{\alpha} x^{\alpha}$ (when $c_{\alpha} \neq 0$) is called a *term of degree $|\alpha|$* of f . The (total) *degree of f* is the maximum degree of the terms in f :

$$\deg f = \max\{|\alpha| : \alpha \in \mathbb{Z}_{\geq 0}^n, c_{\alpha} \neq 0\}$$

The zero polynomial $f = 0$ has no terms and by this degree $-\infty$.

The set of all polynomials in $x_1, \dots, x_n$ with coefficients in k is denoted by $k[x_1, \dots, x_n]$.

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Note: Polynomials are formal expressions, not functions.

Equality of two polynomials is defined through comparison of coefficients.

We make $k[x_1, \dots, x_n]$ an algebraic structure by introducing the basic operations, which are the addition and the multiplication.

Addition:

$$\sum_{\alpha} c_{\alpha} \cdot x^{\alpha} + \sum_{\alpha} d_{\alpha} \cdot x^{\alpha} := \sum_{\alpha} (c_{\alpha} + d_{\alpha}) \cdot x^{\alpha}$$

Multiplication:

$$\left(\sum_{\alpha} c_{\alpha} \cdot x^{\alpha} \right) \left(\sum_{\beta} d_{\beta} \cdot x^{\beta} \right) := \sum_{\alpha, \beta} c_{\alpha} \cdot d_{\beta} \cdot x^{\alpha+\beta} = \sum_{\gamma} \left(\sum_{\alpha, \beta: \alpha+\beta=\gamma} c_{\alpha} \cdot d_{\beta} \right) \cdot x^{\gamma}$$

Remark 1.1.3

We can view k as a subset of $k[x_1, \dots, x_n]$ and by this $k[x_1, \dots, x_n]$ contains $1 \in k$ (The multiplicative identity).

It turns out that $(k[x_1, \dots, x_n], +, \cdot)$ is a unitary, commutative ring:

$$\begin{array}{lll} f + (h + g) = (f + h) + g & f \cdot (h \cdot g) = (f \cdot h) \cdot g & (f + g) \cdot h = f \cdot h + g \cdot h \\ f + g = g + f & f \cdot g = g \cdot f & \\ f + 0 = f & f \cdot 1 = f & \\ f + (-f) = 0 & & \end{array}$$

This holds for all $f, g, h \in k[x_1, \dots, x_n]$. The $-f$ is the unique polynomial whose addition to f gives 0.

Remark 1.1.4

There are three basic choices of n :

$$\begin{array}{l|l|l} n = 1 & x_1 & x \\ n = 2 & x_1, x_2 & x, y \\ n = 3 & x_1, x_2, x_3 & x, y, z \end{array}$$

Later, a connection to geometry will be established. These three basic cases will get connected to line resp. plane and resp. 3-dim. space.

Definition 1.1.5

In algebraic geometry, the space k^n of all n -tuples with the coefficients in k is called the *n -dimensional affine space over the field k* .

From linear algebra we know, that k^n is a k -vector space.

Fun Fact: It's by far not the only space one can consider. Another very common and important space is the projective space. Actually, in the Elliptic Curve Cryptography (ECC), the elliptic curve is considered in the projective space.

Definition 1.1.6

We can evaluate $f = \sum_{\alpha} c_{\alpha} \cdot x^{\alpha} \in k[x_1, \dots, x_n]$ at a point $(a_1, \dots, a_n) \in k^n$ of the affine space obtaining the value.

$$f(a_1, \dots, a_n) = \sum_{\alpha=(\alpha_1, \dots, \alpha_n)} c_{\alpha} a_1^{\alpha_1} \cdots a_n^{\alpha_n}$$

This gives rise to a function $k^n \rightarrow k$. This function is usually denoted again by f , even though $f \in k[x_1, \dots, x_n]$ and $f : k^n \rightarrow k$ is not the same thing. In math, such a situation is called abuse of notation.

I know it's not quite right, but I know what I'm doing

Functions arising this way are called *polynomial functions*.

Remark 1.1.7

In some fields, for example, in finite fields k , different polynomials in $k[x_1, \dots, x_n]$ may define the same polynomial function.

Example

Consider the finite field $\mathbb{F}_2 = \{0, 1\}$ with

+	0	1
0	0	1
1	1	0

and

·	0	1
0	0	0
1	0	1

$$f = x^2 + x \in \mathbb{F}_2[x]$$

$$g = x^3 + x \in \mathbb{F}_2[x]$$

$$f : \mathbb{F}_2 \rightarrow \mathbb{F}_2$$

$a \in \mathbb{F}_2$	$f(a) \in \mathbb{F}_2$
0	0
1	0

$$g : \mathbb{F}_2 \rightarrow \mathbb{F}_2$$

$a \in \mathbb{F}_2$	$g(a) \in \mathbb{F}_2$
0	0
1	0

So, both functions are the same, but the polynomials are different

Fun Fact: Do we need $\mathbb{F}_2[x]$ for anything?

→ Yes! Advanced Encryption Standard (AES) used the field $\mathbb{F}_{256}$, whose construction requires taking a polynomial from $\mathbb{F}_2[x]$.

Theorem 1.1.8

Let the field k be infinite. Then $f \in k[x_1, \dots, x_n]$ is a zero polynomial (all coefficients of f are zero) if and only if $f : k^n \rightarrow k$ is a zero function.